

RECSYS CHALLENGE 2026 • MUSIC-CRS

# State-Driven Retrieval and Learned Re-Ranking for Conversational Music Recommendation

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47,071

tracks in the catalog

11

ways to search

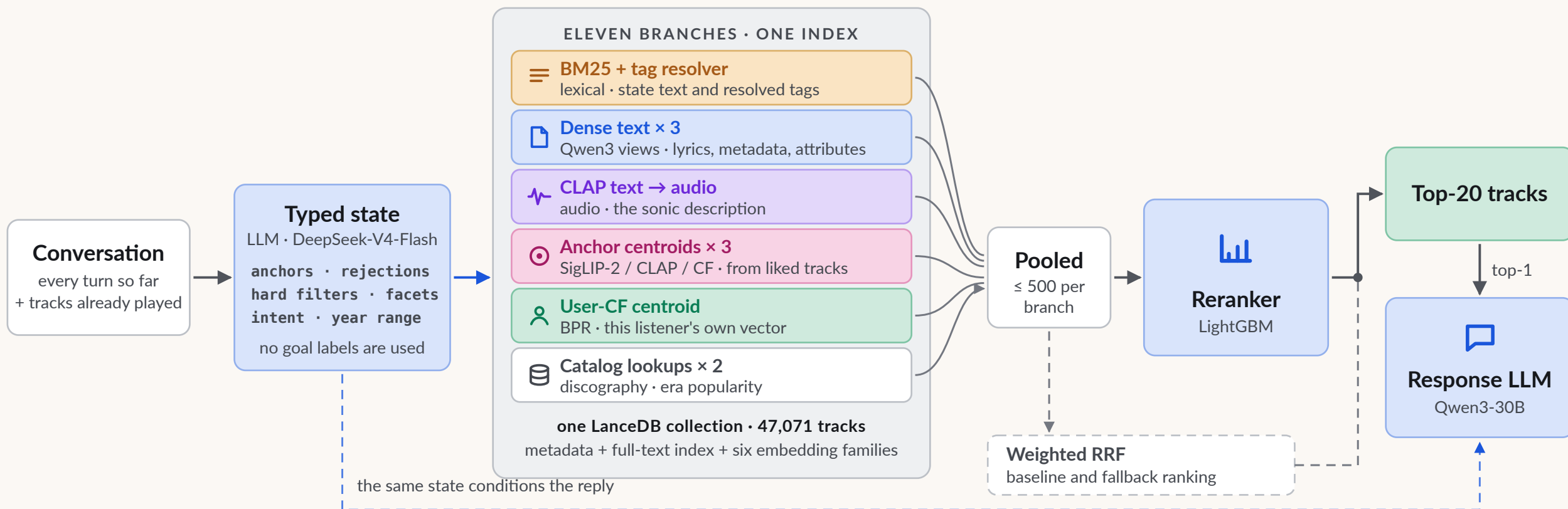
1

learned ranker

A decorative bar chart at the bottom of the slide, consisting of approximately 20 vertical bars of varying heights in a dark gray color.

# ≡ One state, eleven branches, one learned ranker

APPROACH



Organisers supplied the audio (CLAP), cover-art (SigLIP-2), listening-data (BPR) and small text (Qwen3-0.6B) embeddings; we added Qwen3-8B text embeddings and our own bi-encoder.

# Request understanding

APPROACH

WE HAD JUST PLAYED



Ryan Adams & The Cardinals — “Let Us Down Easy”

LISTENER

...more tracks with that kind of emotional depth and acoustic focus, maybe branching out to other singer-songwriter or alt-country artists?

LLM



|              |                                |
|--------------|--------------------------------|
| REQUEST TYPE | new artist                     |
| AVOID        | Ryan Adams & The Cardinals     |
| MOOD         | introspective, emotional depth |
| SOUND        | acoustic-driven                |
| GENRE        | singer-songwriter, alt-country |

The LLM reads **every turn so far**: “other artists” after a Ryan Adams track becomes an artist to avoid, and the mood carries over.



# What the state can hold

APPROACH

## THE REQUEST

- request type
- intent mode
- routing flags

## WHAT THEY LIKE

- anchors and pinned tracks
- per-track feedback
- mentioned entities

## WHAT THEY RULED OUT

- rejected tracks
- rejected artists
- hard filters

## WHAT IT SOUNDS LIKE

- mood facets
- genre facets
- sonic facets

## WHAT IT IS ABOUT

- lyrical theme

## BOUNDS

- release-year window
- process constraints

# Retrieval — eleven branches

APPROACH



×1

## BM25 + tag resolver

keywords and resolved tags  
from the state



×3

## Dense text

Qwen3 embeddings of lyrics,  
metadata, attributes



×1

## CLAP text → audio

the sonic description, matched  
to audio



×3

## Anchor centroids

near liked tracks: cover art,  
audio, listening



×1

## User-CF centroid

this listener's own BPR vector



×2

## Catalog lookups

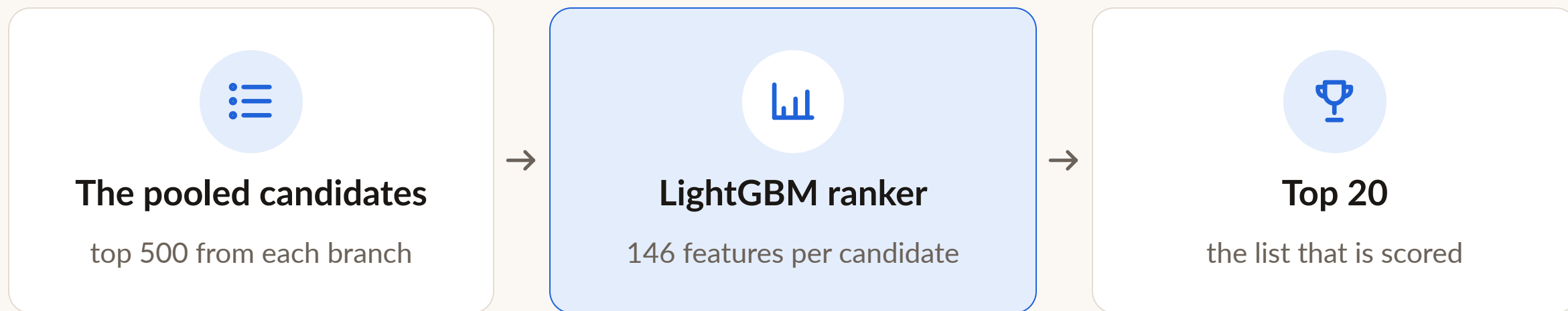
the artist's discography; era  
popularity

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Each branch returns its **top 500**; the ranker sees their union.

# Reranking

APPROACH





# What the ranker leans on

APPROACH



Share of the ranker's split gain, by feature family. One signal — the bi-encoder cosine — carries **59%**; constraints and rejections carry **0.7%**.



# Response generation

APPROACH

LISTENER

...it feels a bit too serious. I'm looking for something more uplifting and positive, really something to lift my spirits.

OUR REPLY



“Perfect — ‘Happy Up Here’ by Röyksopp is exactly the kind of uplifting, cheerful electronic track you’re looking for. With its bright synths and hopeful groove, it’s designed to lift your spirits and make you smile.”

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A second LLM (**Qwen3-30B**) writes one short reply about the top track, from its tags and the state.

## ✓ How it scored

RESULTS

| OFFICIAL CODABENCH SPLIT    | NDCG@20 | CAT.   | LEX.   | JUDGE | COMPOSITE |
|-----------------------------|---------|--------|--------|-------|-----------|
| Blind-A · development phase | 0.4380  | 0.0313 | 0.8028 | 4.70  | 0.5799    |
| Blind-B · final             | 0.2537  | 0.0315 | 0.7862 | 3.30  | 0.3811    |

NDCG@20

**-42%**

0.4380 → 0.2537

JUDGE (1-5)

**-1.40**

4.70 → 3.30

DIVERSITY TERMS

**flat**

catalog and reply-wording  
diversity barely moved

Blind-B also withholds the conversation-goal annotations — but the deployed ranker never used them at inference, so that is not the explanation.



# Label example — “from a different artist”

WHAT IS NOT IDEAL

RECOMMENDER AGENT PLAYED



Bonobo — “Cirrus”

LISTENER, NEXT TURN

...something with a similar chill, electronic vibe, maybe with a subtle beat, but **from a different artist?**

ANSWER KEY FOR THAT TURN



Bonobo — “Jets”

MOVES TOWARD GOAL

✗ Our judges: wrong — same artist

NEXT TURN

“Jets” is cool, but I was actually hoping for something **from a different artist this time...**

ANSWER KEY



Bonobo — “Pieces”

MOVES TOWARD GOAL

✗ Our judges: wrong — same artist

Twice asked to leave Bonobo; both answer keys are Bonobo, **marked helpful**.



# What the relabelling found

ANALYSIS

## 17%

of training turns keep the artist the listener had just asked to leave

18,222 turns — both judges or the arbiter marked the label as artist anchoring

## 113,393

turns re-judged

every labelable train and dev turn

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Retrained on these labels, the ranker scored lower **against the original labels** — a test that could never favour the fix — so we kept the originals.



# Why we trained on the dev set

LESSONS



**1 paid LLM call per turn**  
to read each turn into a state



**Too costly for 121.6k turns**  
so no features on the training set



**Trained on the 8,000 dev turns**  
the same turns we checked it on

Dev, turns it trained on



0.384

Blind-B, official



0.254

nDCG@20. The dev row is an earlier model version, shown because it is the number we selected on. Source: paper §4.3, §5.3.



# Extracted, but the ranker did not act on it

WENT WRONG



*"I gotta break away from **Deltron** and **Del** completely"*

**Dead Prez — Hip Hop**

state held both artists as avoid

NOT ACTED ON



*"branching out to **other** singer-songwriter artists"*

**Ryan Adams — When The Stars Go Blue**

state held Ryan Adams as avoid

NOT ACTED ON



*"still focusing on **Juanes**... haven't found the depth I'm after"*

**Juanes — Me Enamorado De Ti**

state held Juanes as rejected

NOT ACTED ON

We wanted the ranker to use what the state recorded; it served that artist anyway. We think the labels taught it to: they often repeat the just-played artist, so rejections carried only **0.7%** of its decisions.

# No data from outside the catalog

WHAT IS NOT IDEAL

## A tempo and a key

*“exactly 126.70 bpm, in G minor”*



the catalog has no tempo or key field  
**no branch can match on either**

## A soundtrack

*“a specific 80s synth-pop song from the **Breaking Bad** soundtrack”*



we served Soft Cell — “Tainted Love”  
**no film or TV links in the catalog**

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The state was right both times. The gap is **knowledge, not understanding**.



# The reply was never checked

WHAT IS NOT IDEAL

The track is not in the catalog, so retrieval could not return it — the reply presents our substitute as if it were the request

1

*"I wanted 'Watercolors' by Pat Metheny specifically."*

"Got it — here's 'Alfie' by Pat Metheny..."

NOT FLAGGED

2

*"Could you please play Sarah McLachlan's 'Fallen'?"*

"You're in the right place — 'Hold On' by Sarah McLachlan is a perfect match..."

NOT FLAGGED

Our pick broke the listener's constraint — the reply recommends it anyway

3

*"...something from a **different artist** this time"*

"...while this track is by him, it's from a different era and band..." — Ryan Adams again

NOT FLAGGED

4

*"...we're still focusing on **Juanes**..."*

"I know you're looking to explore beyond Juanes, but 'Me Enamoré De Ti' is a standout..."

NOT FLAGGED

One pass, no check: the reply could flag **neither unavailable tracks nor unmet constraints**. The prompt was chosen on Blind-A (judge 4.70 there, 3.30 on Blind-B).



# Everything is released

THANK YOU



Paper, poster, slides, code, models, data

[npatta01.github.io/music-crs-2026](https://npatta01.github.io/music-crs-2026)